AKASH S

Embedded Firmware Engineer | IoT Systems

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SUMMARY

Embedded Firmware Engineer with hands-on experience designing and deploying production-grade firmware for STM32 (ARM Cortex-M) and ESP32 platforms. Skilled in C programming, low-level protocol driver integration (UART, SPI, I2C), DMA-driven memory transfers, and OTA firmware updates. Proven track record of architecting reliable MQTT communication pipelines and developing industrial IoT tracking systems. Seeking to leverage expertise in hardware-software integration and bare-metal fundamentals to drive scalable firmware solutions.

TECHNICAL SKILLS

Languages: Embedded C, C, Python, Data Structures & Algorithms (C)

Microcontrollers & Hardware: STM32 (ARM Cortex-M), ESP32, Bare-Metal Driver Development, CMSIS, HAL Architecture

Peripherals & Protocols: UART, SPI, I2C, DMA, Interrupts (NVIC), Timers/PWM, GPIO, MQTT, HTTP, GSM/GPRS

Firmware Concepts: Over-the-Air (OTA) Updates, Memory Mapping, Register Manipulation, State Machines, Circular Buffers

Developer Tools & Hardware Lab: ST-Link, STM32CubeIDE, Logic Analyzers, Oscilloscopes, Git, GitHub, VS Code

Secondary / Edge Technologies: Raspberry Pi, Jetson Nano, OpenCV/YOLOv8, Linux (Ubuntu), CCNA Networking Basics

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | KritiLabs Technologies Pvt Ltd *May 2025 – Present | Chennai, India*

- Architected and deployed secure Over-the-Air (OTA) firmware update mechanisms for STM32-based industrial trackers and electronic lock systems, eliminating the need for manual on-site flashing.
- Engineered non-blocking UART communication drivers using circular DMA buffers to interface with GSM/GPRS modules, maintaining zero packet loss during continuous MQTT data transmissions.
- Configured and optimized low-level hardware protocols (I2C, SPI, UART) for sensor data acquisition and peripheral communication across custom PCB hardware.
- Developed cloud-driven automation workflows for ESP32 edge devices using lightweight MQTT and HTTP protocols.
- Conducted hardware-level debugging and signal validation using oscilloscopes and logic analyzers to troubleshoot timing issues, hardware interrupts, and bus conflicts.

Embedded & AI Trainee | Innospace Automation Pvt Ltd *Jun 2023 – Sept 2024 | Chennai, India*

- Developed an edge-based industrial computer vision system utilizing YOLOv8 to automate real-time forklift entry detection and enhance warehouse safety monitoring.
- Designed real-time image processing and data pipelines, achieving a 95% anomaly detection accuracy in operational environments.
- Integrated edge AI models with cloud infrastructure, optimizing execution speed and improving data processing efficiency by 98%.
- Collaborated with cross-functional engineering teams to integrate embedded vision solutions into existing industrial automation frameworks.

ACADEMIC PROJECTS

Sri Sairam Engineering College | Sep 2021 – Apr 2025

- Multimodal Smart Glove for people with disabilities — secured USD 5,050 in funding through IEEE EPICS.
- Smart Spectacles for visually impaired users — real-time obstacle detection using machine learning and image processing; submitted as an IEEE conference paper.
- AI-Based QR-Enabled Pharmaceutical Dispenser for healthcare automation — published at an IEEE international conference.
- Crop Prediction and Continuous Monitoring System using IoT sensors and analytics.

EDUCATION

B.E., Electronics and Communication Engineering

Sri Sairam Engineering College, Chennai | Sep 2021 – Apr 2025 | CGPA: 7.34

HONORS, PUBLICATIONS & CERTIFICATIONS

- **IEEE EPICS Grant Recipient ($5,050 USD):** Secured competitive international funding for the design and deployment of an IoT-enabled Multimodal Smart Glove.
- **IEEE Conference Author:** Published research paper on *"AI-Based QR Enabled Pharmaceutical Dispenser"* in IEEE Xplore.
- **IEEE Conference Author:** Submitted paper on *"Smart Spectacles for Real-Time Obstacle Detection using Embedded Machine Learning"* to IEEE.
- **Core Technical Certifications:** CCNA (Modules 1–3) | Raspberry Pi & Hardware Integration (EC-Council) | C & Python Programming (Cisco / Spoken Tutorial).